

Chain-of-Thought (CoT) Logic Engine

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1. Introduction

Large Language Models (LLMs) are capable of answering complex questions, but they can sometimes make logical mistakes or hallucinate information. Prompt Engineering helps improve their performance by providing structured instructions. One effective technique is **Chain-of-Thought (CoT) Prompting**, which encourages the model to solve problems through logical reasoning before producing a final answer.

This project focuses on designing a reasoning prompt that forces the AI to analyze a problem carefully, verify its reasoning, and perform a self-correction step before delivering the final response.

2. Objective

The objective of this project is to design a Chain-of-Thought prompt that improves reasoning accuracy by forcing the AI to think systematically, verify its logic, and minimize hallucinations. The prompt should guide the model through multiple reasoning steps while ensuring the final answer is both correct and reliable.

3. What is Chain-of-Thought Prompting?

Chain-of-Thought Prompting is an advanced prompting technique where the AI is instructed to solve problems through multiple logical steps instead of directly generating an answer.

Benefits include:

- Better reasoning
- Reduced hallucinations
- Higher accuracy

- Improved transparency
- Easier error detection
- Better performance on complex tasks

4. Project Requirements

According to the project guidelines, the prompt should:

- Solve problems step by step.
- Avoid skipping logical steps.
- Perform self-correction.
- Verify calculations.
- Reduce hallucinations.
- Provide the final answer only after verification.

5. Master Prompt

You are an expert reasoning AI.

Your task is to solve every problem carefully using logical reasoning.

Instructions:

1. Read the question carefully.
2. Break it into smaller logical steps.
3. Solve each step one at a time.
4. Explain your reasoning clearly.
5. Check every assumption and calculation.
6. Perform a self-correction phase:
 - Review your reasoning.
 - Identify any mistakes.
 - Correct them if necessary.
7. After verification, provide the final answer.

Rules:

- Never guess.
- If information is missing, clearly state it.
- Prioritize correctness over speed.

6. Test Cases

Test Case 1

A train travels 60 km in one hour.

How long will it take to travel 210 km?

The train's speed is 60 km/h.

Use the formula:

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$\text{Time} = \frac{210}{60} = 3.5 \text{ hours}$$

So it will take 3.5 hours, which is the same as 3 hours and 30 minutes.

Check:

- In 3 hours, the train travels $3 \times 60 = 180$ km.
- In another 30 minutes (0.5 hour), it travels $0.5 \times 60 = 30$ km.
- Total distance: $180 + 30 = 210$ km ✓

Final answer: 3.5 hours (3 hours 30 minutes).

Test Case 2

Sara has twice as many books as Ali.

Together they have 36 books.

How many books does each have?

Let Ali have x books.

Since Sara has **twice as many** books as Ali, Sara has $2x$ books.

Together they have 36 books:

$$x + 2x = 36$$

$$3x = 36$$

$$x = 12$$

So:

- Ali has 12 books.
- Sara has 24 books.

Check:

- Sara has twice as many as Ali: $24 = 2 \times 12 \checkmark$
- Total: $24 + 12 = 36 \checkmark$

Final answer:

- Ali: 12 books
- Sara: 24 books

Test Case 3

A farmer needs to cross a river with a wolf, a goat, and a cabbage.

The boat can carry only the farmer and one item.

If the wolf is left alone with the goat, the wolf eats the goat.

If the goat is left alone with the cabbage, the goat eats the cabbage.

How can the farmer get everything across safely?

A safe sequence is:

1. **Take the goat across** the river.
 - Left behind: wolf and cabbage (safe).
2. **Return alone.**
3. **Take the wolf across.**
 - On the far side: wolf and goat.
 - **Bring the goat back** to avoid leaving it alone with the wolf.
4. **Take the cabbage across.**
 - Left behind: goat (safe by itself).
 - Far side: wolf and cabbage (safe).
5. **Return alone.**
6. **Take the goat across** one final time.

Now all three—the **wolf**, **goat**, and **cabbage**—are safely on the other side.

Final crossing order:

1. Goat →
2. Farmer ←
3. Wolf →
4. Goat ←
5. Cabbage →
6. Farmer ←
7. Goat →

7. Results

The structured prompt consistently produced better reasoning compared to a normal prompt. The self-correction phase reduced logical errors and improved confidence in the final answers.

9. Conclusion

This project demonstrated how Chain-of-Thought prompting significantly improves the reasoning capabilities of Large Language Models. By forcing the AI to think step by step and review its own reasoning before producing a final answer, the quality and reliability of responses increased. The self-correction stage helped identify potential errors and reduced hallucinations, making the prompt suitable for solving complex logical problems.

10. Learning Outcomes

After completing this project, I learned:

- How Chain-of-Thought prompting works.
- How to create structured reasoning prompts.
- How self-correction improves AI responses.
- How to reduce hallucinations.
- How to design prompts for complex logical reasoning.
- How to evaluate prompt quality through testing.